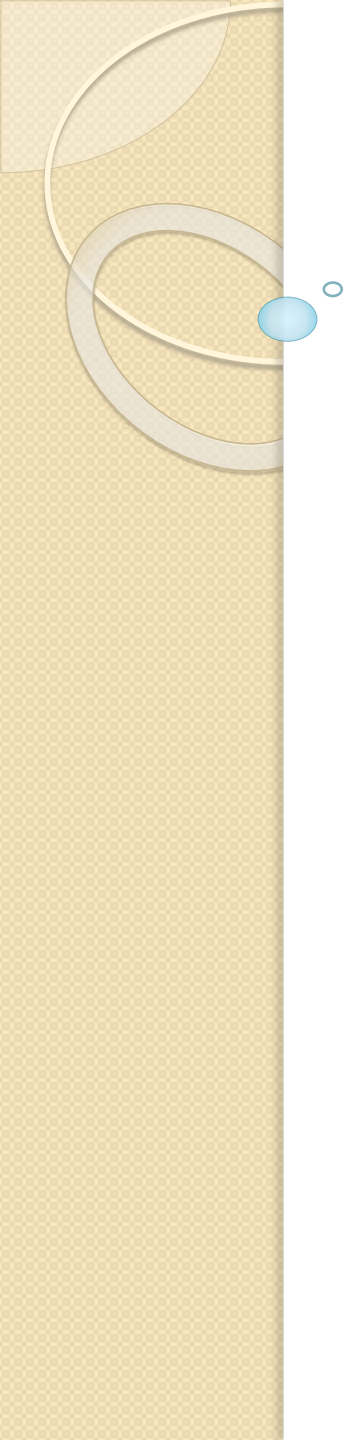




Chapter-5

Physical Database Design

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Contents

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- Physical Database Design Process in Relational Databases
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 - Types of single level ordered index
 - Multilevel indexes
 - Dynamic multilevel indexes using B-Trees & B+ Trees
 - Indexes on multiple indexes
 - Query Optimization

Revision

Conceptual model vs Logical model vs Data model

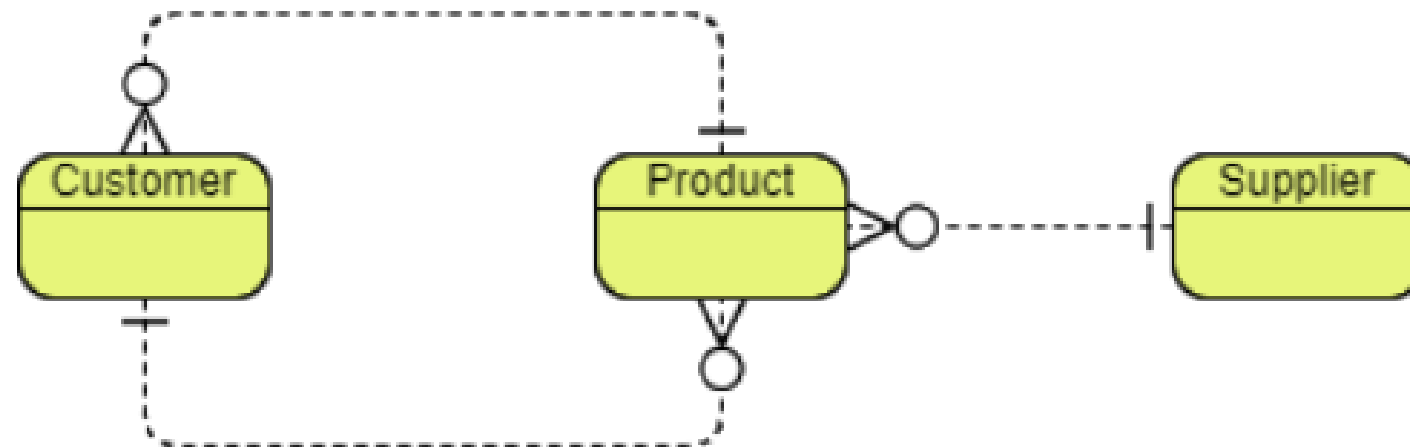
- An ER model is typically drawn at up to three levels of abstraction:
 - Conceptual ERD / Conceptual Data Model
 - Logical ERD / Logical Data Model
 - Physical ERD / Physical Data Model

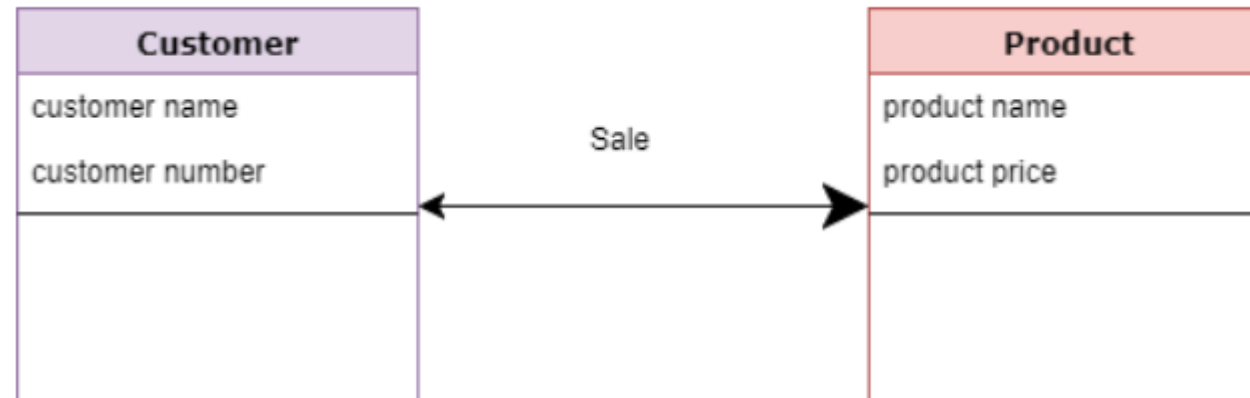
Revision

cont...

Conceptual model

- Conceptual ERD **models** the **business objects** that should exist in a system and the relationships between them.
- A conceptual model is developed to present an **overall picture of the system** by recognizing the business objects involved. It defines what entities exist, NOT which tables.





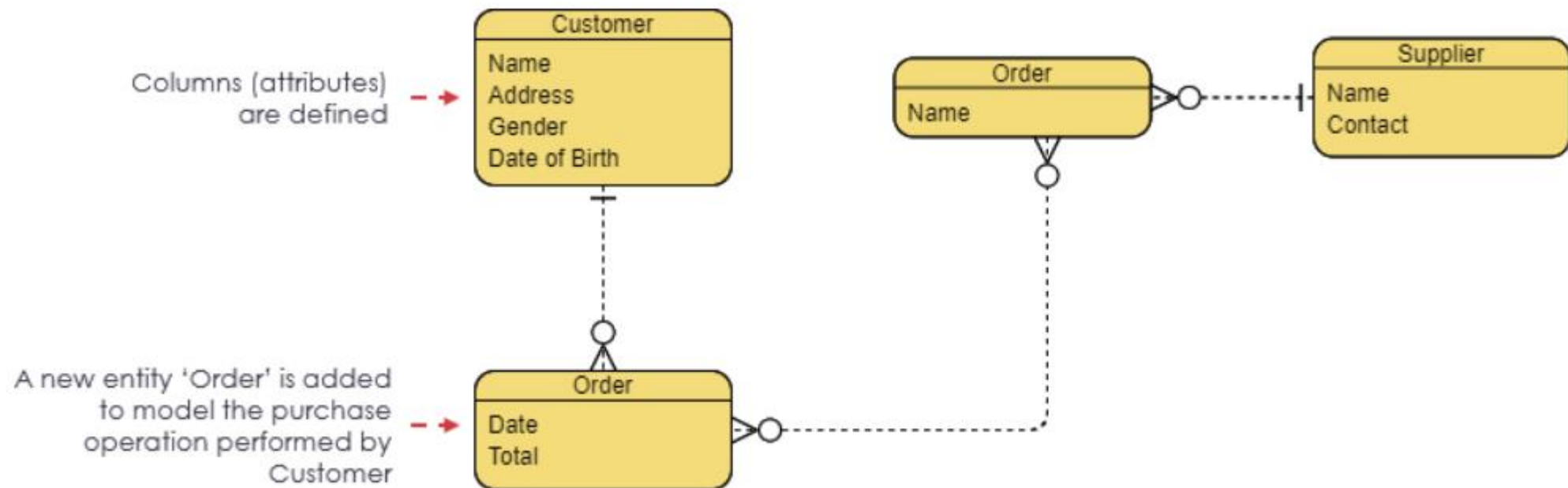
Conceptual Data Model

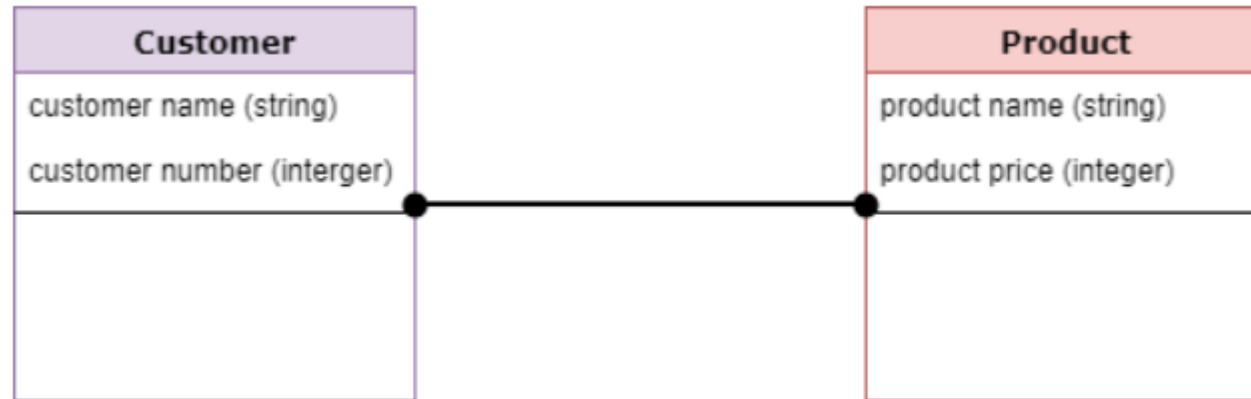
Revision

cont...

Logical ERD

- Logical ERD is a **detailed version** of a Conceptual ERD.
- A logical ER model is developed to enrich a conceptual model by **defining explicitly the columns** in each entity and introducing operational and transactional entities.





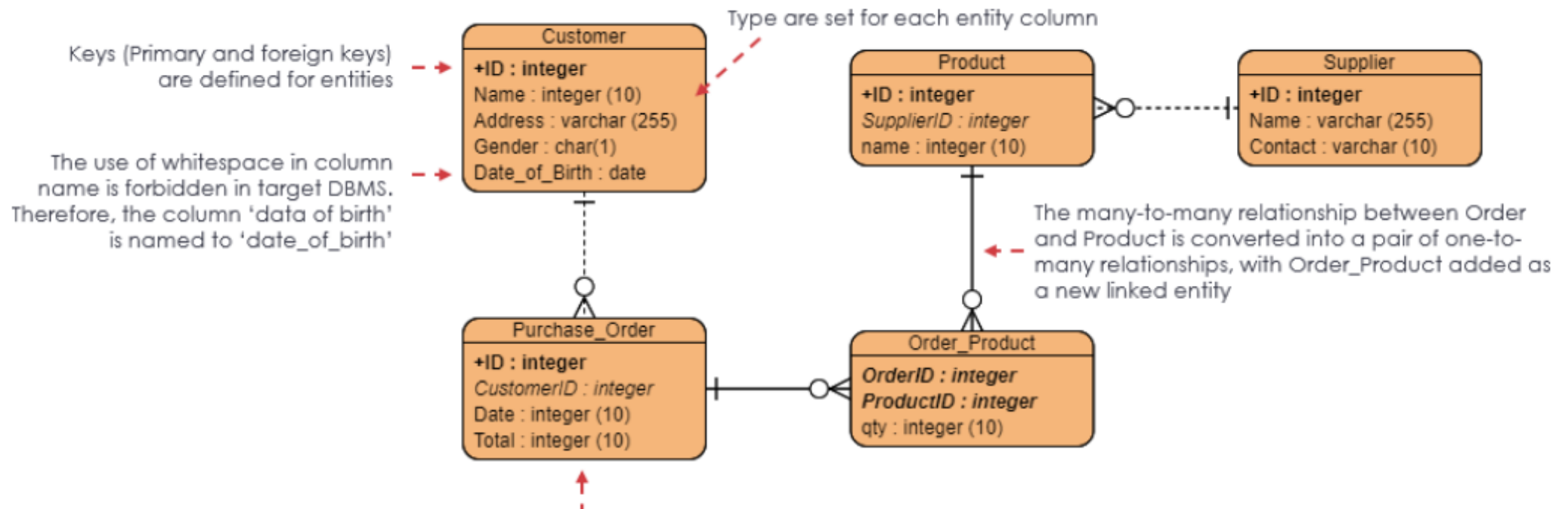
Logical Data Model

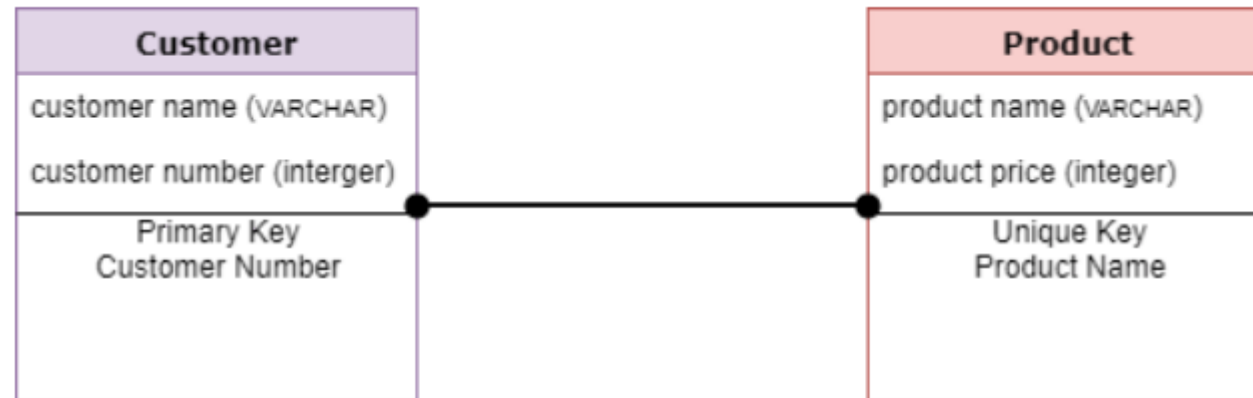
Revision

cont...

Physical ERD

- It represents the **actual design blueprint of a relational database**. It elaborates on the logical data model by assigning each **column** with **type, length, nullable**, etc.
- Since a physical ERD **represents how data should be structured and related in a specific DBMS** it is important to consider the convention and restriction of the actual database system in which the database will be created.
- Make sure the **column types** are **supported** by the **DBMS** and **reserved words** are **not used** in naming entities and columns.





Physical Data Model

Revision

cont....

Summary

ERD feature	Conceptual	Logical	Physical
Entity (name)	Yes	Yes	Yes
Relationship	Yes	Yes	Yes
Column		Yes	Yes
Column's Type		Optional	Yes
Primary Key			Yes
Foreign Key			Yes

Introduction

- Physical database design is the **process of transforming logical data models** into **physical data models**.
- **Its Purpose** is to **translate** the **logical description** of data **into technical specifications** for **storing** and **retrieving** data.
- Information **needed** for **physical file** and **database design** includes:
 - **Normalized relations** plus their size estimate
 - **Expectations** and **requirements** for **response time**, **data security**, **backup & recovery**, **retention** and **integrity**
 - Descriptions of **where** and **when data** are **used**, **entered**, **retrieved**, **deleted**, **updated**, and **how often**
 - **Descriptions** of the **technologies** used to **implement** the database
 - **Definitions** of **attributes** in the tables

Physical Database Design Process: Inputs

- ✓ **Tables produced by logical database design**
 - ✓ **Normalized tables**
- ✓ **Business environment requirements**
 - ✓ **Response time:** delay from the time that the enter key is pressed to execute a query and the **result appears on the screen.**
 - ✓ **Throughput:** measure of **how many queries from simultaneous users** must be **responded** in a **given period of time** by the **application** and the **database** that supports it.
- ✓ **Data characteristics**
 - ✓ **Data volume assessment:** how much data will be in the database, **roughly how many records is each table expected to have?**
 - ✓ **Data volatility:** refers to **how often stored data is updated, deleted?...**

Cont...

- ✓ **Application characteristics**
 - ✓ Which **applications** are the most **important** to the company?
 - ✓ What is the **nature of applications** that will use the data?
 - ✓ Which **data will be accessed** by each application?
 - ✓ Application data requirements
 - ✓ Application priorities

- **Operational Requirements**

- **Security** : protecting data from **theft** or malicious destruction and making sure that **sensitive data** is **accessible only** to those who are concerned.
- **Backup & recovery**:
 - **Backup**: copy of the entire data to ensure database safety.
 - **Recovery**: being **able to recover** a table or database that has been **corrupted** or **lost** due to **hardware** or **software failure**.

- **Hardware and Software Characteristics**

- **DBMS Xcs**: Attribute **data type** options, and **SQL query** features, which must be known and taken into account during physical database design.
- **Hardware Xcs**: **Processor speeds** and **disk data transfer rates**.

Things to be analyzed by Database Designers cont...

- Database queries & transactions
- The expected frequency of invocation of queries & transactions
- The time constraints of queries & transactions
- The expected frequencies of update operations
- The uniqueness constraints on attributes

Decisions

- By considering the above inputs and analyzing them, the decisions to be done on the physical database design are;
 - Attribute data types
 - Physical record and data/file storages
 - File organizations
 - Indexes
 - Query optimization

Inputs

- Normalized relations
- Attribute definitions
- Response time expectations
- Data security needs
- Backup/recovery needs
- Integrity expectations
- DBMS technology used

Leads to

Decisions

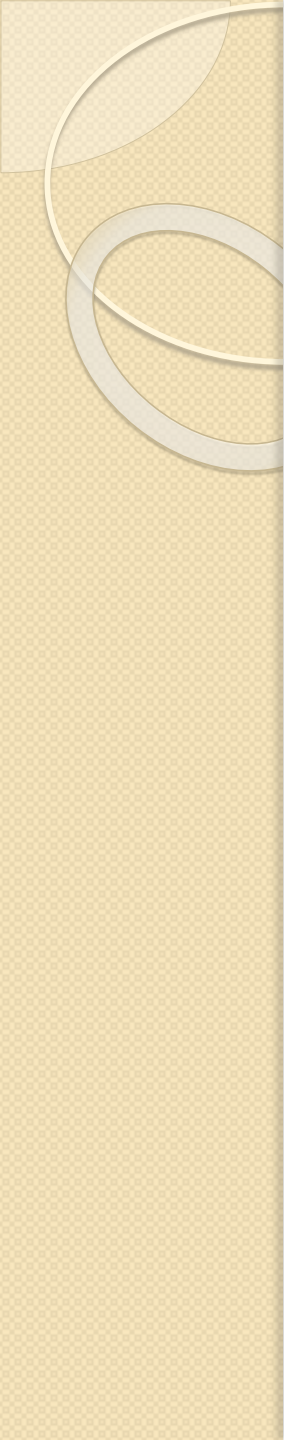
- Attribute data types
- Physical record descriptions (doesn't always match logical design)
- File organizations
- Indexes and database architectures
- Query optimization

Choosing Attribute Data Types

- CHAR - fixed-length character
- VARCHAR() - variable-length characters
- NUMERIC - positive/negative numbers
 - Exact
 - SMALLINT → 16 bit
 - INT → 32 bit
 - LONG → 64 bit
 - DECIMAL → 128 bit ...
 - Approximate
 - FLOAT → 32 bit with $\approx$ 6-9 digit precision...
 - REAL → 32 bit with $\approx$ 15 digit precision...
 - DOUBLE → 64 bit with $\approx$ 32 bit precision...

Physical Records and Data/File Storage

- **Physical record:** is a group of fields stored in adjacent memory locations and retrieved together as a unit whenever necessary.
- **Page:** the amount of data read or written in one input/output (I/O) operation.
- **Blocking factor:** number of physical records per page.
- The collection of data that makes up a computerized database must be stored physically on some computer storage medium.
- The database management software (**DBMS**) allow users to create, retrieve, update, delete and process this data whenever needed.



Data/File Storage: Categories

Three main storage categories:

- Primary storage
- Secondary storage
- Tertiary storage

Con't...

Primary storage

- It includes storage media that store a data which can be operated directly by the computer central processing unit (CPU).
- Primary storage **includes**;
 - The computer main memory (**RAM**) and
 - Smaller but faster **cache memories**.
- Primary storage usually provides **fast read** and **write** to data. It is of **limited storage capacity**.
- Primary storage devices are more **expensive**.
- The **contents** of main memory are **lost** in case maybe when power failure, a system crash or other issue occurs.

Cont...

Secondary storage

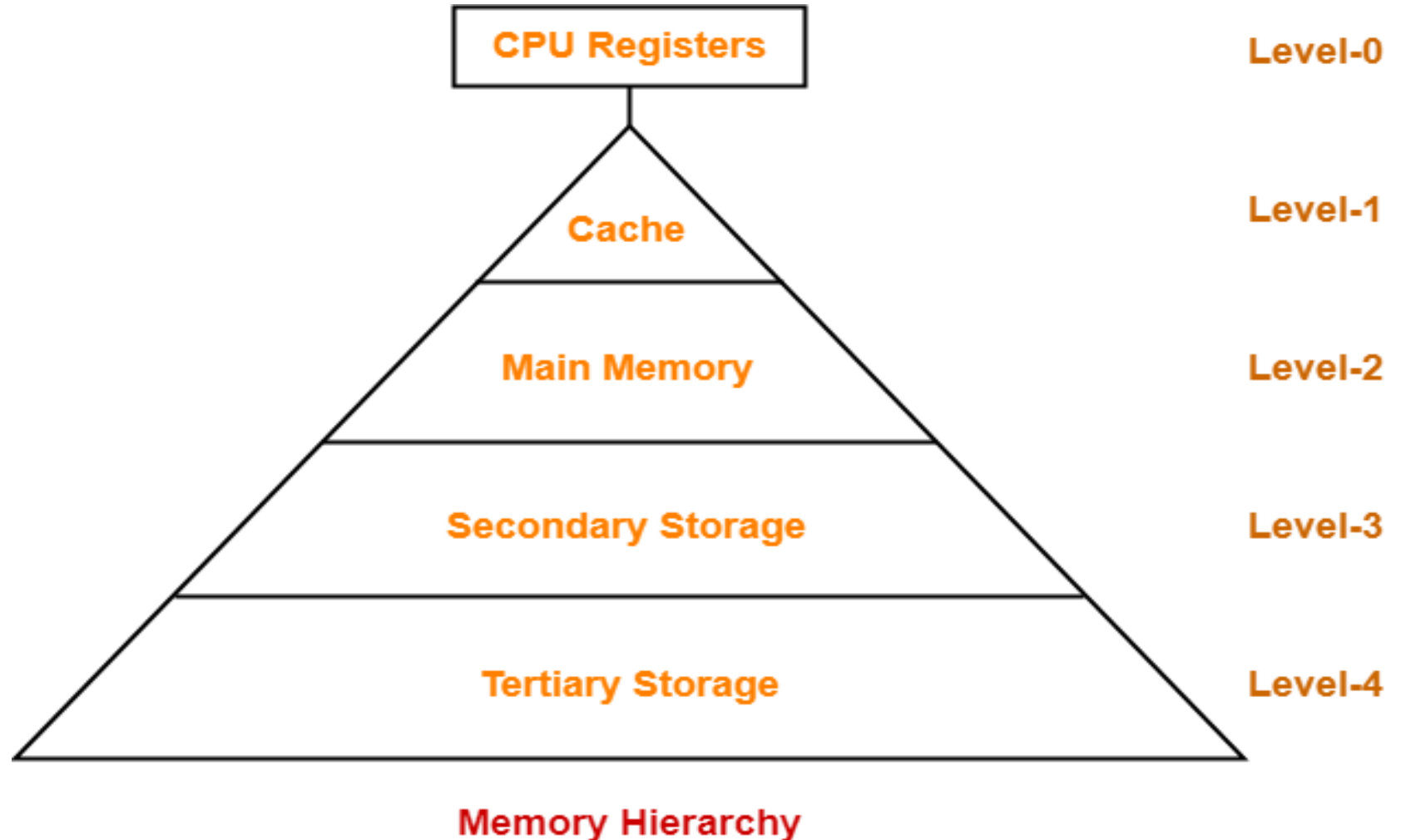
- Includes large storage devices such as computer hard disk (**HDD**) and **SSD**.
- These devices usually have
 - A larger capacity,
 - Less cost, and
 - Provide **slower access** to data than primary storage devices.
- Data in secondary storage **cannot be processed directly** by the **CPU**, it must first be copied into primary storage.
- They are **called online storage devices** because they **can be accessed in short period of time** whenever needed.

Tertiary storage

- **Optical disks** (CD-ROMs, DVDs, and other similar storage media) and **magnetic tapes** which are **removable** media are used in today's systems as **offline storage** for **archiving databases**.

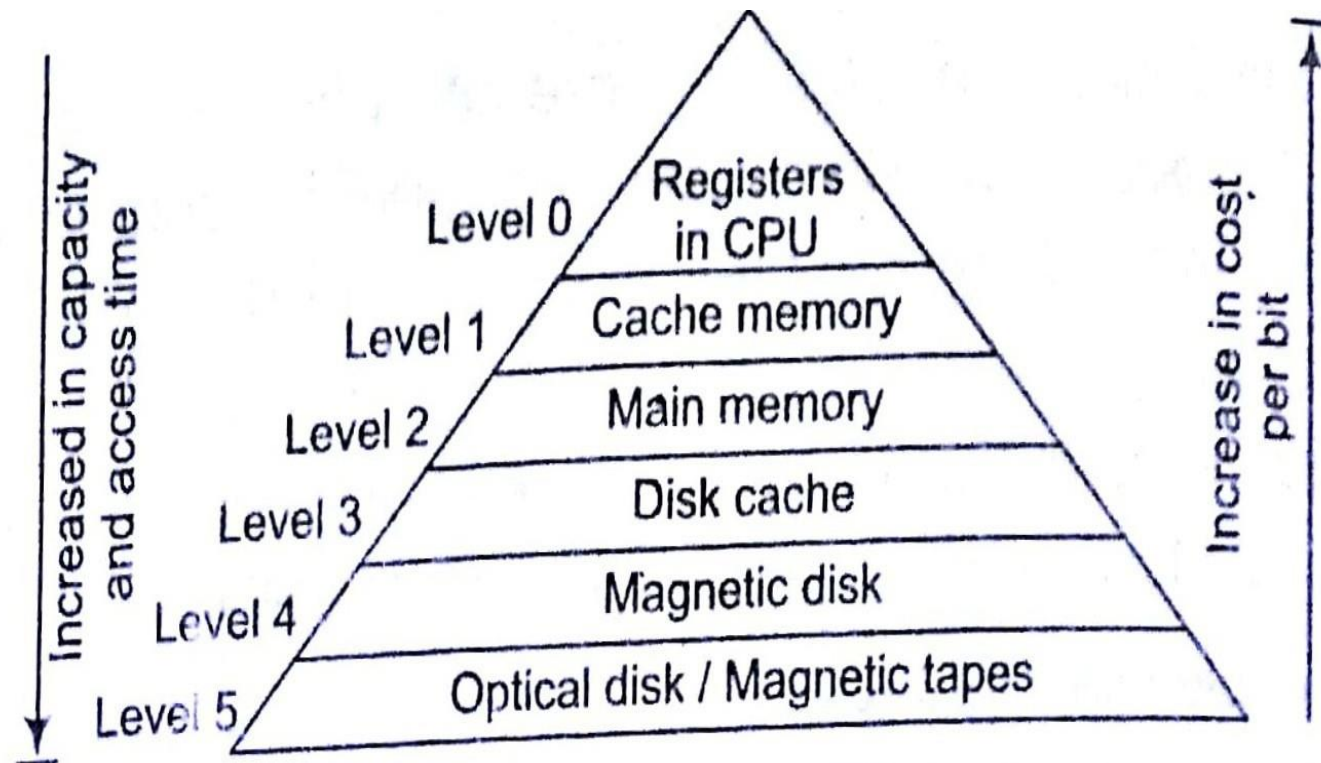
Memory Hierarchies and Storage Devices

Memory hierarchies



Cont...

- Storage capacity, access speed and cost comparison, as we move from one level to the other



Data Storage in Databases

- Data stored in databases
 - **Persistent data**
 - **Temporary data**
- Databases typically store **large amounts** of data that must persist **over long periods** of time.
 - This is the reason that the data is often referred to as **persistent data**.
 - Parts of this data are **accessed** and **processed repeatedly**.
 - Parts of the data may be **accesses** and **processes very rarely**.
- **Transient/temporary** data: persists for **only a limited time** during program execution.

Cont...

- Typical **database applications need** only a **small portion** of the database at a time for **processing**.
- When a specific **operation** on **data** is **needed**, first the **data must be located on disk**→ **copied** to **main memory** to be ready for processing → **processed**→ **re-written** to the disk if its content is changed.
- The **data stored** on **disk** is **organized as files** of **records**.
- Each record is a collection of data values that can be interpreted as facts about **entities**, their **attributes** & their **relationships**.
- Records **should be stored** on disk in a manner that makes possible to locate records/data **efficiently** when they are needed.

Operations on files

Operations on files are usually grouped into:

- **Retrieval operations** → do not change any data in the file, but only locate certain records so that their field values can be examined and processed.
- **Update operations** → change the file by insertion or deletion of records or by modification of field values.
- In either case, we may **have to select one or more records** for retrieval, deletion, or modification based on a selection condition (or filtering condition) → which specifies a criteria that the desired record or set of records must satisfy.

Cont...

- Each **located record** in the given file/table **is checked** to **determine** whether it **satisfies** the full **selection condition**.
- Actual **operations** for locating and accessing file records may **vary from system to system**.
- Typically, high-level programs, such as **DBMS software programs**, access records by using certain commands, such as:
- Open, Reset, Find/Locate, Read/Get, FindNext, Delete, Modify, Insert, Close, Scan....

Cont...

- **Open**→ prepares the file for reading or writing.
- **Reset**→ sets the file pointer of an open file to the beginning of the file.
- **Find/ Locate**→ searches for the first record that satisfies a search condition.
- **Read/Get**→ copies the current record from the buffer to a program variable in the user program.
- **FindNext**→ searches for the next record in the file that satisfies the search condition.
- **Delete**→ deletes the current record and (eventually) updates the file on disk to reflect the deletion.

Cont...

- **Modify** → modifies some field values for the current record and (eventually) updates the file on disk to reflect the modification.
- **Insert** → inserts a new record in the file by locating the block where the record is to be inserted, transferring that block into a main memory buffer (if it is not already there), writing the record into the buffer, and (eventually) writing the buffer to disk to reflect the insertion.
- **Close** → completes the file access by releasing the buffers and performing any other needed cleanup operations.
- **Scan** → if the file has just been opened or reset, scan returns the first record; otherwise it returns the next record.

Cont...

- At this point, it is worthwhile to note the difference between the terms file organization and access method.
- **File organization**→ refers to the organization of the data or a file into records, blocks, and access structures, which includes the way records and blocks are placed on the storage medium and interlinked.
- **Access method**→ provides a group of operations that can be applied to get/retrieve a file.

File Organizations

- There are several primary file organization types, which determine how the file records are physically placed on the disk.
 - **Heap file (unordered file)** → places the records on disk in no particular order → by appending new records at the end of the file.
 - **Sorted file (sequential file)** → keeps the records ordered by the value of a particular field (called the sorting key).
 - **Hashed file** → uses a hash function applied to a particular field (called the hash key) to determine a record's placement on disk.
- Secondary organization or auxiliary access structures → **allows efficient access & storage** to file records based on alternate fields than those that have been used for the **primary file organization**.

Decisions to be done: Indexing (remember slide 14)

- *Indexes* are a powerful tool used in **the background of a database to speed up querying**.
- Indexes power queries by providing a method to quickly lookup the requested data.
- Simply put, an **index** is a **pointer** to data in a table.
- **Design decisions about indexing**
 - **Whether to index an attribute:** the general rules for creating an index on an attribute are that
 - The **attribute must either be a key**, or
 - There **must be some query** that **uses that attribute** either in a **selection** or a **join** condition.
 - **What attribute or attributes to index on:** an index can be **constructed** on a **single attribute**, or on more than one attribute if it is a composite index.

Design decisions about indexing

Con't...

- **Whether to set up a clustered index:** At most, **one index** per table can be a **primary** or **clustering** index.
 - In most RDBMSs, this is specified by the keyword `CLUSTER`. (If the attribute is a **key**, a **primary index** is **created**, whereas a **clustering index** is **created** if the attribute is not a key.)
 - If a table requires **several indexes**, the decision about **which one should be the primary or clustering index** depends upon whether keeping the table ordered on that attribute is needed.

Cont...

- Design decisions about indexing
 - **Whether to use a hash index over a tree index:** RDBMSs can use B+ trees for indexing.
 - **Whether to use dynamic hashing for the file:** for files that are very volatile → those that grow and shrink continuously, one of the dynamic hashing schemes would be suitable.

Index Types

- Index
 - Dense
 - Sparse
 - Single Level
 - Multilevel
- Single level ordered indexes
 - Primary index,
 - Secondary, and
 - Clustering
- Multilevel indexes
 - Search tree
 - B -tree
 - B+ -tree

Dense Vs Sparse Index

Dense :

- An **index entry** is **created** for **every search key value** (for each records) in each block.
- This index **contains** search **key value** and a **pointer** to the actual record.
- Large **index size**
- **Less time** needed to locate arbitrary data.

Sparse :

- **One index entry** for **each block**.
- Indexes are **created** only for **some** of the **data records**.
- **Small index size**
- **More time needed** to locate arbitrary data
- Records must be **clustered** or **arranged** in blocks
- **Faster write**

Cont...

Data

first_name	last_name	email	phone
Velvet	Clyde	v.clyde@company.biz	...
Oscar	Nathan	o.nathan@company.biz	...
Susan	Tobias	s.tobias@company.biz	...
Kiley	Hardy	k.hardy@company.biz	...

Page 1

Rachelle	Debra	r.debra@company.biz	...
Valerie	Raleigh	v.raleigh@company.biz	...
Flynn	Gilbert	f.gilbert@company.biz	...
Isabel	Brenna	i.brenna@company.biz	...

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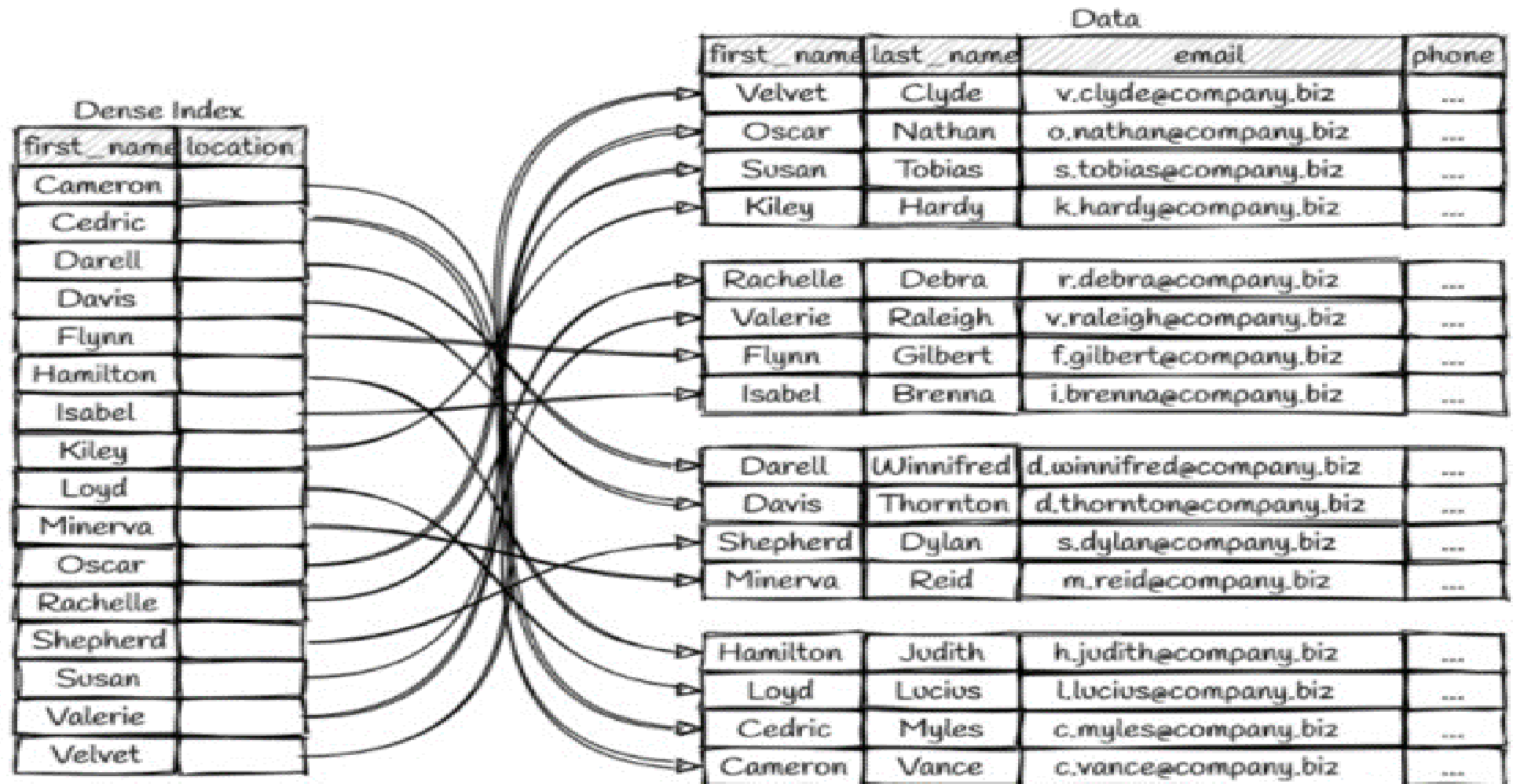
Darell	Winnifred	d.winnifred@company.biz	...
Davis	Thornton	d.thornton@company.biz	...
Shepherd	Dylan	s.dylan@company.biz	...
Minerva	Reid	m.reid@company.biz	...

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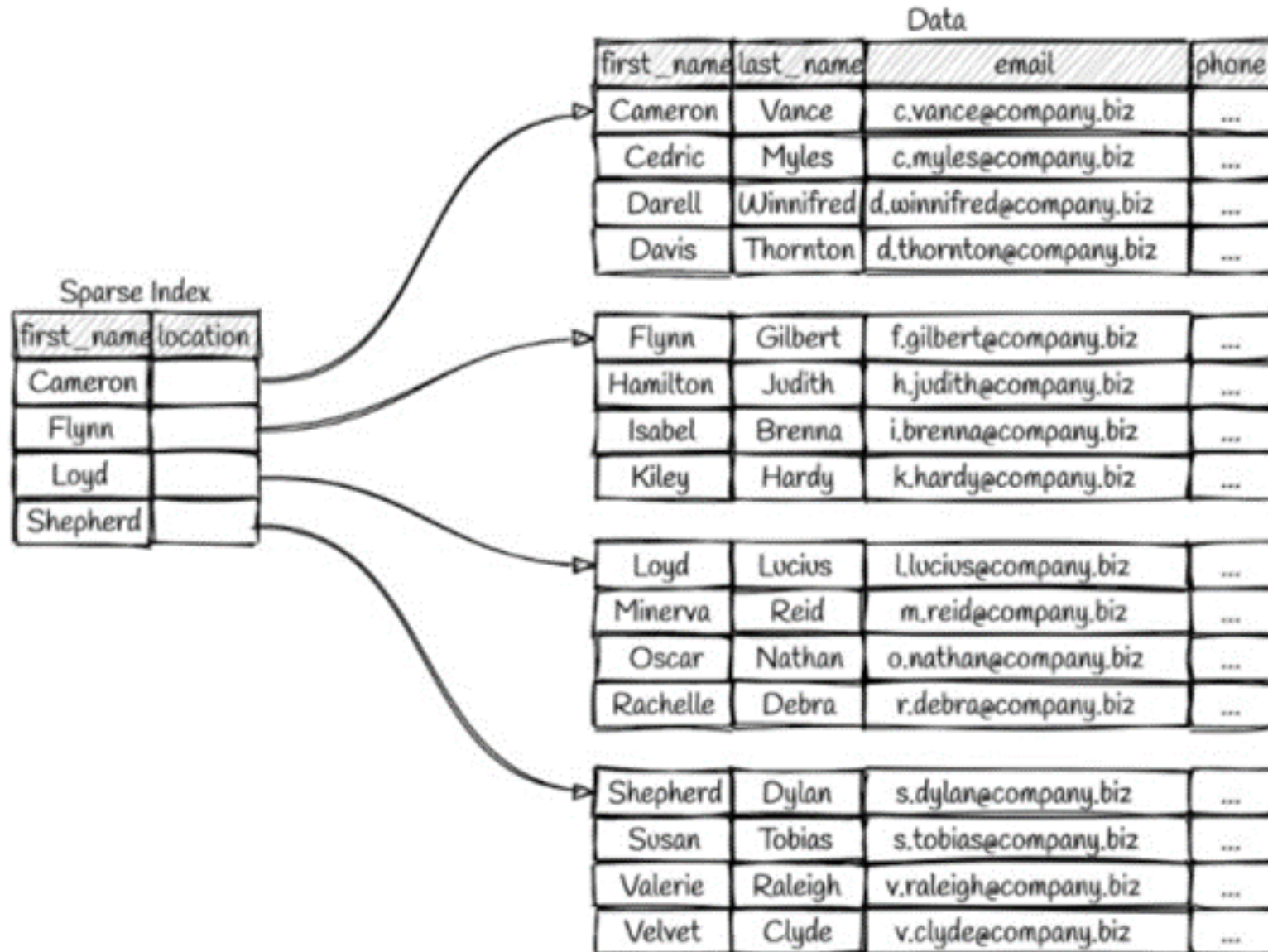
Hamilton	Judith	h.judith@company.biz	...
Loyd	Lucius	l.lucius@company.biz	...
Cedric	Myles	c.myles@company.biz	...
Cameron	Vance	c.vance@company.biz	...

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Dense Indexing

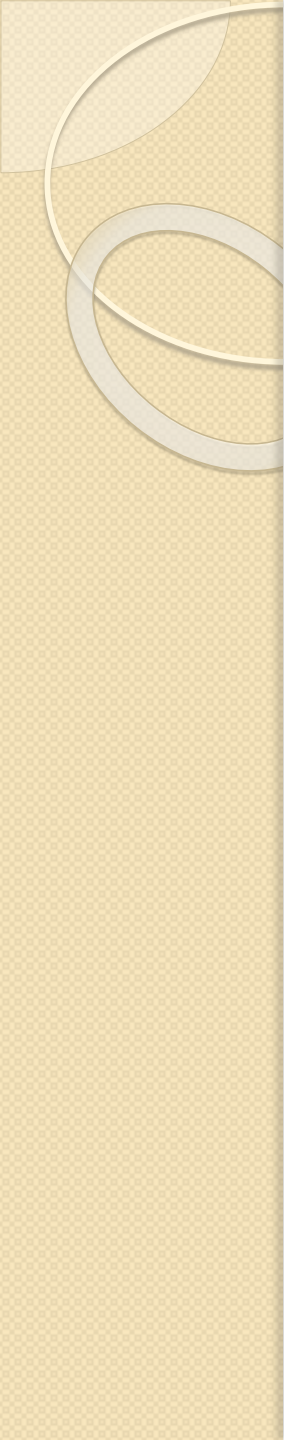


Sparse Indexing



Reading assignment (Have at least a highlight understanding about the following)

- Primary index
- Clustering index
- Secondary index
- Multilevel Index (Search tree, B –tree, B+ -tree)
- Dynamic multilevel indexes using B-Trees & B+ Trees
- Indexes on multiple indexes
- Query Optimization



Thank you !!!